# WSO2 Integrator: MI 4.6.0 Training – Prerequisites

## About this document

Every trainee needs the setup below installed and tested **at least 1 day before Day 1**. The training runs for 4 days (7 hours a day) and covers WSO2 Integrator: MI 4.6.0, from Developer Fundamentals to selected Developer Advanced topics.

The trainer provides the installers that are hard to find in a GitHub repo (see *Shared resources*).

## Laptop, access and network

On the last day, MI, VS Code, ActiveMQ, MySQL and the backend run at the same time.

| Item | Requirement |
|---|---|
| OS | macOS or Linux |
| Internet | Needed for the VS Code extension, connectors and Gmail (Day 2) |
| Free local ports | 8290, 8253, 9164 (MI) · 9090, 9091 (Hospital backend) · 61616, 8161 (ActiveMQ) · 3306 (MySQL) |

## Core software (all 4 days)

Install these four items on every laptop. MI 4.6.0 is tested with JDK 21 and 25, so use **JDK 21**.

| Software | Version | Notes | Download |
|---|---|---|---|
| Java (JDK) | 21 | OpenJDK, Temurin, Corretto or Oracle JDK. Set `JAVA_HOME`. The Hospital backend also runs on JDK 21. | Tested JDKs |
| WSO2 Integrator: MI | 4.6.0 | Extract the pack; this folder is `MI_HOME` | Install MI |

| Software | Version | Notes | Download |
|---|---|---|---|
| VS Code + WSO2 Integrator: MI extension | Latest | Used for every lab. Also builds the class mediator on Day 4, so no separate Maven is needed. | Install the extension |
| cURL | Any | Used to test APIs. Postman is optional. | curl.se , Postman |

## Additional software and accounts

The labs also need a message broker, a database and a few tools. Items marked MI_Training repo can be downloaded from github.com/veneerac/MI_Training.

| Item | Version | Setup notes | Where to get it |
|---|---|---|---|
| Apache ActiveMQ Classic | 5.19.1 (MI supports Classic up to 6.1.x) https://activemq.apache.org/components/classic/download/classic-05-19-01 | Copy the client JARs from `ACTIVEMQ_HOME/lib` to `MI_HOME/lib` : activemq-broker, activemq-client, activemq-kahadb-store, hawtbuf, slf4j-api and activeio-core (in `lib/optional` ). Start it with `bin/activemq console` (macOS/Linux) or `bin\activemq start` (Windows). | MI_Training repo, or activemq.apache.org |
| MySQL Server | **8.0 or later** (MI 4.6.0 is tested with MySQL 8 and 9) | Install it and note the root password. It's used for data services and the API user store. | mysql.com/downloads |
| MySQL JDBC driver (Connector/J) | 8.2.0 | Copy `mysql-connector-j-8.2.0.jar` to `MI_HOME/lib` . Driver class: `com.mysql.cj.jdbc.Driver` . | MI_Training repo |
| Hospital backend service | 2.0.0 | Run `java -jar Hospital-Service-JDK11-2.0.0.jar` (port 9090). It's the sample backend used in the mediation labs. | MI_Training repo only. The link in the public tutorials no longer works. |
| MI CLI ( `mi` ) | 4.6.0 | Extract it and add the `bin` folder to `PATH` | MI CLI |
| Integration Control Plane (ICP) | Matching MI 4.6.0 | Used on Day 3 to monitor MI | ICP |
| Gmail account | — | Turn on 2-step verification and create an **app password** for the email connector lab. Test SMTP/IMAP access from your network. | Google account |

## What each day uses

ActiveMQ is needed from Day 1, the Hospital backend from Day 2, and MySQL from Day 3. Everything must be ready before Day 1.

| Day | Topic | Labs | Needs, beyond the core software |
|---|---|---|---|
| Day 1 | Basics and triggering messages | 1–5: setup, REST API, proxy service, scheduled tasks · 6: event listener (JMS) | ActiveMQ (Lab 6) |
| Day 2 | Mediation | 7: routing · 8: transformation · 9: orchestration · 10: templates · 11: store and forward · 13: email connector | Hospital backend (Labs 7–11) · ActiveMQ (Lab 11) · Gmail app password (Lab 13) |
| Day 3 | Data, quality and operations | 15: data services · 16: debugging · 17: unit tests · 18: CLI and ICP | MySQL + Connector/J (Lab 15) · MI CLI + ICP (Lab 18) |
| Day 4 | Advanced | 01: JMS producer and consumer · 02: custom mediator · 05: Secure Vault · 07: Basic Auth for APIs · 08: VFS (optional) | ActiveMQ + Hospital backend (Lab 01) · MySQL + Connector/J + Hospital backend (Lab 07) |

## Shared resources

The trainer shares a `Shared Resources` folder with three items and the ActiveMQ client JARs. Download it from github.com/veneerac/MI_Training before the training.

| Folder | File | Notes |
|---|---|---|
| ActiveMQ | `apache-activemq-5.19.1-bin.tar.gz` | Official Apache release (SHA-512 checked). macOS and Linux. Windows users download `apache-activemq-5.19.1-bin.zip` from activemq.apache.org. |
| MySQL JDBC Driver | `mysql-connector-j-8.2.0.jar` | From Maven Central (SHA-1 checked). Works with MySQL 8.0 and later. |

| Folder | File | Notes |
| --- | --- | --- |
| Hospital Backend | `Hospital-Service-JDK11-2.0.0.jar` | Sample backend for the labs, tested on JDK 21. The `source` folder holds its code. |

The repo's `ActiveMQ/client-jars-for-MI_HOME-lib/` folder holds the 6 ActiveMQ client JARs, taken from the pack: activemq-broker, activemq-client, activemq-kahadb-store, hawtbuf, slf4j-api and activeio-core. Copy those 6 JARs and `mysql-connector-j-8.2.0.jar` to `MI_HOME/lib`, then restart MI. These are the only third-party JARs the labs need. MI 4.6.0 already ships with the JMS API and the MySQL scripts in `MI_HOME/dbscripts/mysql`.